

ProLM: a plasma proteomics pretrained model for the general population

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Motivation: From Genetic Risk to Plasma Proteins

Polygenic Risk Score (PRS)

- PRS (polygenic risk score): sum of risk alleles across the genome associated with specific phenotypes.
- **Limitation:** considers only genetic risk, not environmental factors.

In contrast, plasma proteins capture

- genetic predisposition
- environmental exposure
- lifestyle
- aging

Proteomic-based risk scores are challenging:

- Difficult to interpret complex regulatory networks.

Datasets

UK Biobank data

Pre-training dataset

- 2,873 plasma proteins from 15,499 individuals (with Imputation by 'miceforest')
- Free of chronic disease at baseline and throughout follow-up
- unlabeled

Fine-tuning dataset

- Individuals who developed one of 16 chronic diseases
- 1,723 healthy individuals
- labeled

External Validation dataset

- China Kadoorie Biobank (CKB)

ProLM Architecture

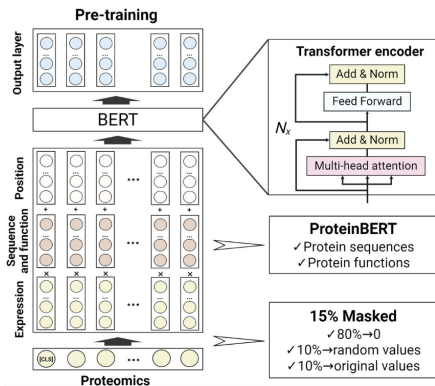


Fig 1. Pre-training Architecture

Input

- 15,499 individuals $\times$ 2,873 protein expression values

Protein representation

- Elementwise product of individual protein expression & ProteinBERT embedding

Encoding

- 2,873 proteins $\rightarrow$ 6 chunks
- [CLS] added to each chunk
- CLS pooling $\rightarrow$ global feature

PPI-Biased Self-Attention

Standard Self-Attention + PPI Bias

- Standard transformer Q, K, V
- PPI(matrix captures evidence of protein–protein association
- 7 categories of evolutionary and functional evidence
- Confidence score in $[0, 1]$
- Trainable weight controls the contribution of the PPI bias

Pre-training Objective

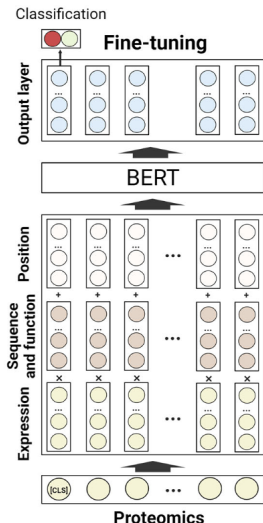
Masking

- 80% → set to zero
- 10% → replaced by a random value
- 10% → unchanged

Objective

- Reconstruct masked protein expression values
- Optimized using MSE loss

Fine-tuning for Disease Prediction



Prediction task

- Separate binary classifier for each of 16 chronic diseases

Fine-tuning

- Unfreeze only the final 8 BERT layers
- Add a fully connected classification head

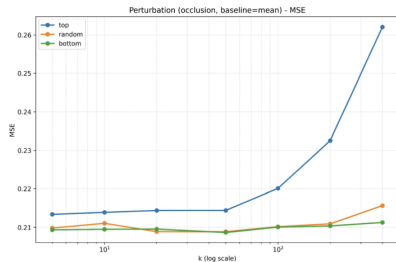
Training

- Cross-entropy loss

Fig 1. Disease prediction fine-tuning

Analysis of Pre-training

Protein Importance (IG)



Supplementary Fig 2. Top-ranked proteins show greater performance degradation under perturbation (MSE).

Demographic Representation

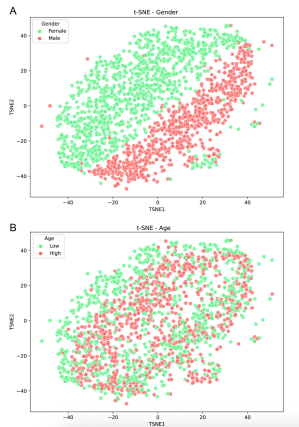


Fig 2. Learned representations capture demographic structure.

Fine Tuning: Disease Prediction Performance

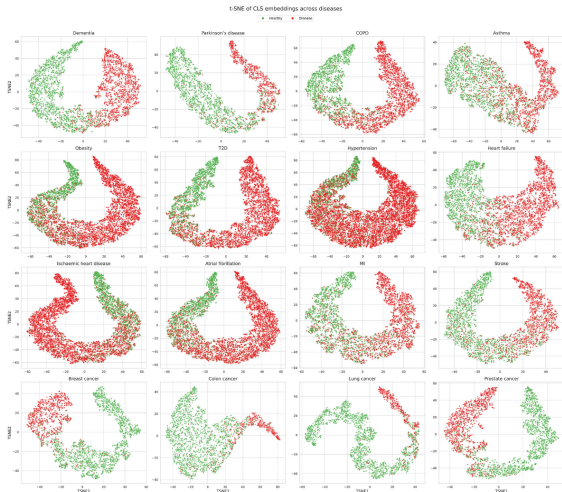


Fig 3. t-SNE visualization of proteomic embedding.

Comparison

- MLP, FT-transformer, ResNet, LightGBM, CatBoost
- DeLong test for AUC differences

Fine Tuning: Clinical Evaluation of ProRS

1. Kaplan–Meier Analysis

- ProRS-based risk groups:
Low (< 0.33), Medium ($0.33\text{--}0.66$), High (> 0.66)
- Distinct outcome trajectories across the three risk groups

2. Cox Proportional Hazards Model

- Compare ProRS with clinical predictors: Age+Sex, ASCVD, and PANEL
- Evaluate combinations of ProRS + clinical predictors
- Discrimination assessed by Harrell's C-index

3. Calibration and decision curve analysis

- Compare predicted disease risk with observed event rates: ProRS with clinical predictors Age+Sex, ASCVD, and PANEL

Fine Tuning: Clinical Evaluation of ProRS

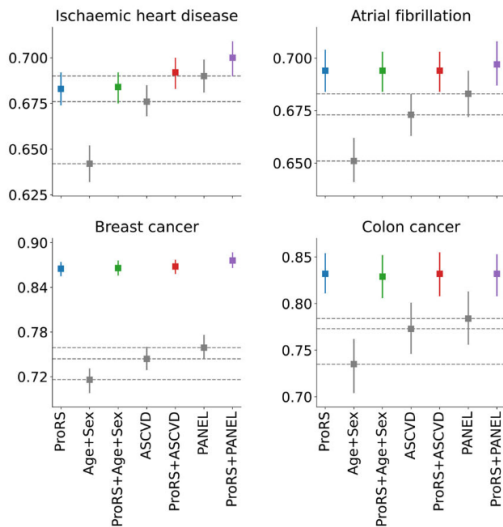


Fig 5.A Comparison of ProRS and clinical predictor models (only 4 of 16 figures)

Key finding

- Adding clinical predictors provided **little additional gain**
- Combined models were not consistently superior to ProRS alone

Interpretation

- Proteomic risk may already capture substantial clinical risk information

Model Interpretation: Genetic / Longitudinal trajectory

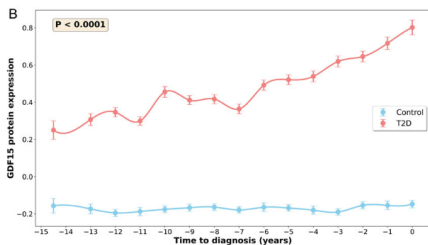


Fig 7. Longitudinal trajectories of GDF15 expression

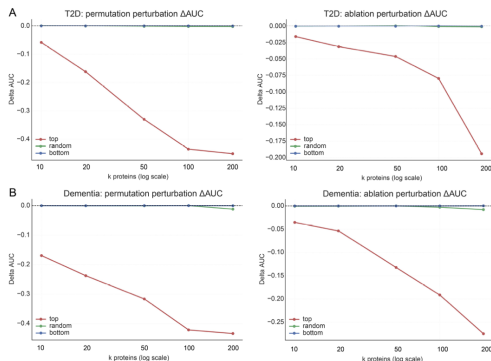
Longitudinal trajectory

- Altered protein levels before diagnosis
- Monotonic temporal patterns
- Different trajectories between cases and controls

Biological relevance

- Consistent with known disease biology / GWAS
- e.g., GDF15, REN, GFAP

Model Interpretation: Validating Protein Importance



Supplementary Fig 12. Performance under grouped protein perturbation

Integrated Gradients

- Rank proteins by global attribution score

Perturbation validation

- Top / random / bottom proteins
- Permutation and ablation
- Larger performance drop for top-IG proteins

Fine Tuning: Disease Subtype Classification

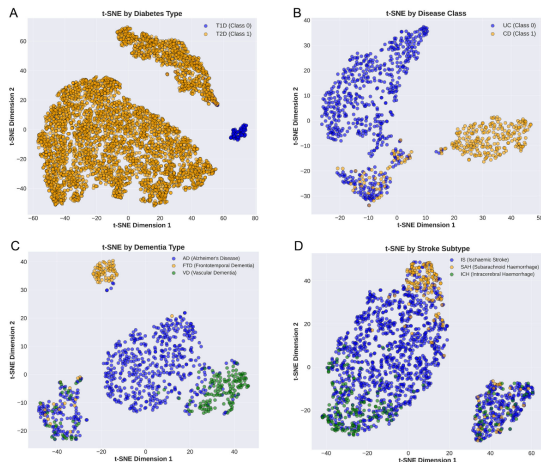


Fig 8. Subtype classification performance for representative diseases

Tasks

- Dementia, Stroke: multi-class
- Diabetes, IBD: binary

Fine-tuning

- Pre-trained ProLM + classification head
- Cross-entropy loss

Model Reduction Test

Supplementary Table 13. Ablation study.

Ablation study	Loss	Accuracy	MSE	MAE	R²
ProLM	0.2109	0.8509	0.2109	0.2894	0.5283
No-ProteinBERT	0.2158	0.8487	0.2168	0.2931	0.5229
No-PPI	0.2224	0.8436	0.2224	0.2970	0.5036

Reduction settings

- Remove ProteinBERT embedding
- Remove PPI matrix in the attention mechanism
- Evaluate on the pretraining model

External Validation in CKB

Disease	UKB Validation		CKB External Validation	
	Accuracy	AUC	Accuracy	AUC
MI	0.7798	0.8641	0.6997	0.7901
CHD	—	—	0.6533	0.7677
COPD	0.8052	0.9203	0.9097	0.7634
Hypertension	0.8347	0.7918	0.8291	0.7339
Stroke	0.8000	0.8789	0.8570	0.7415
T2D	0.8767	0.9448	0.9362	0.8221

Supplementary Table 5, 14. Evaluation metrics of the disease prediction task

Experiment: U-CAN Pan-Cancer Proteomics

U-CAN cohort

- 1,477 patients
- 12 cancer types
- Plasma collected at diagnosis, before treatment
- Olink PEA proteomics
- 1,463 proteins measured by NPX

Preprocessing

- Long-format Olink data $\rightarrow$ individual $\times$ protein matrix
- Median imputation for missing expression values
- UniProt accession $\rightarrow$ amino-acid sequence
- 6 proteins without a unique usable sequence excluded

$1,477 \times 1,457$ expression matrix

Experiment: Classification Performance

Validation Performance

Metric	Score
Accuracy	0.7500
Macro Precision	0.7692
Macro Recall	0.7823
Macro F1	0.7685
Macro OvO AUC	0.9736

Table. Best Validation Performance (Epoch 21)

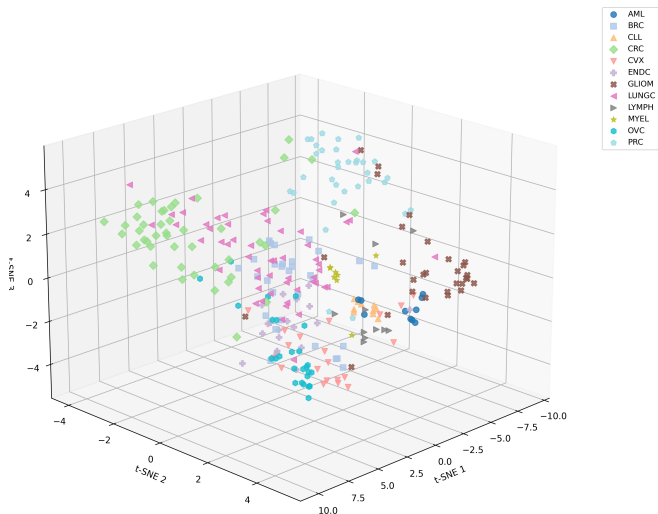
Task

- 12-class cancer classification.
- Final 8 BERT layers fine-tuned with cross-entropy loss

Multiclass evaluation

- 12 cancer types
- 66 pairwise One-vs-One AUC
- Macro averaging across pairs

Experiment: Learned Cancer Representation



Observation

- Separation across subtypes
- Dispersed cancer representations show poorer classification performance.

Fig. t-SNE of validation-set global CLS representations